Contents

1 Spatial concepts **1**

1.1 The ‘core model’ of geographical economics 1

1.1.1 The core model’s story 1

1.1.2 The core model 2

List of Figures **1**

List of Tables **1**

Bibliography **11**

Chapter 1

Spatial concepts

1.1 The 'core model' of geographical economics

1.1.1 The core model's story

The central story of the **core model** builds on two older stories about trade. Each of these tells its story by thinking about two regions and asking, what leads them to trade with each other? What do they gain by doing so? Before the monopolistic competition era, this trade was theorised to happen because each region has its own unique set of factors. Each would thus benefit by some trade, even if one actually had access to both internally; this is the central idea of comparative advantage, and had remained fundamentally the same since Ricardo first formulated it.

Empirical reality, however, appeared not to support this idea. Evidence showed trade taking place intra-industry; for example, two regions both selling each other very similar cars, or apparently similar intermediate factors (Brakman and Heijdra 2001 p.27). The **Constant Elasticity of Substitution (CES)** function offers a very simple way to explain this phenomenon, as section **??** has already introduced. If two regions both contain consumers with a **love of variety**, having access to a variety of cars to buy will increase utility overall. Krugman has a simple example. Two regions in autarky (not trading) each build three types of car, and their consumers thus gain from that variety - able to access those three. What happens if trade is opened up? Here, some additional assumptions can enter into the simple picture to explain the outcome. Firstly, the total amount of labour across both regions does not change: the number of workers is always the same. Second, each of the car industries - six in total in both regions - has increasing returns. The more workers each has, the more efficient their output. So what happens if each isolated region moves to open trade with their neighbour? One possibility is as follows: the highest utility for both regions comes about by each region producing two varieties. Four in total are now made, each at a higher output efficiency. That story contains a number of crucial steps. First, opening trade restructures the economy of both regions. In this case, one car-maker in each region disappears. Secondly, no exogenous difference between the regions is required to show how trade can be beneficial. Thirdly, it highlights the important job that increasing returns

is playing: any firm making a single variety of car will have no incentive to make anything other than its own. If it did, it would lose the efficiency of larger-scale production. Each is thus 'locked in' to its own variety.

double-check everything I'm saying here with the ch.2 monocomp section.

So: Krugman's trade theory (1979) showed how trade patterns seen in the real world can come about with no fundamental initial differences.

> "In standard trade models location does not matter, because when transport costs are absent there is nothing to gain from being in specific location" (Brakman and Heijdra 2001 p.27).

Krugman's later **core model** introduced this one extra simple factor: transport costs between the two regions. The impact of this change, however, was dramatic. It demonstrated a mechanism: as transport costs dropped, one of these two regions would enter a positive feedback and become the productive 'core' containing all manufacturing workers, relegating the other to the 'periphery'. Just slight differences in initial conditions determine which is which. This is not a new idea in itself: Myrdal first talked of 'cumulative causation' (see e.g. Acs and Varga 2002 p.137). Krugman's stated aim was to formalise this in a neoclassical equilibrium setting. And to lure economists...

The **core model** also showed it is possible for this outcome to occur 'catastrophically': a marginal change in transport costs can trigger the start of a feedback process that leads to a core emerging.

The issue of realism of assumptions will be discussed at length throughout the thesis, and is the particular focus of chapter **??**. It is useful to note that a central purpose of the **core model** was to demonstrate it is mathematically *possible* for a core-periphery structure to emerge endogenously. It is, seen this way, meant to illustrate that such a dynamic is mathematically possible. That is some way from saying that it is either economically possible, or that the dynamic actually exists in reality. But treating it is an existence proof helps in understanding why certain model choices were made. For example, the choice of a Cobb Douglas function

That phrase. Keeping things separate, unaffected by cost changes

Makes it much easier to produce a model where the scaffold can be stripped away to reveal, as cleanly as possible, the essential dynamic. This is seen in **Geographical Economics (GE)** as a strength of the model, not a weakness, though various commenters have criticised this or that assumption. Could do with listing some of those.

1.1.2 The core model

Overview of section

This section gets into the workings of the **core model** in detail. Two main sources are used. First, Brakman *et al*. 2009, which contains a detailed explanation. Brakman is also used by

Fowler to build his agent version. The second source is Fujita *et al*. 2001. As will be explained, this takes a slightly different approach (continuous rather than discrete goods) but provides much useful extra information. There are a number of other outlines of the **core model** in the literature including, of course, Krugman's own in his original paper (Krugman, 1991). These will be used to clarify certain points. To keep terminology and choice of parameter names consistent, those from Brakman *et al*. are used, so where quotes are given from Fujita *et al*. 2001, any mathematical labels have been changed to match. While various additions and extensions to the **core model** have been developed over time (and indeed Brakman *et al* say a number of these have "gained the reputation of being a 'core' model of **GE**", p.134), Krugman's original model structure is the basis for the **core model** this thesis discusses.

The usual approach to describing the **core model** is to follow the logical path of the mathematical argument, beginning at the micro level and building up towards the model's conclusions. While this is perfectly sensible mathematically, it is not a particularly good way to open the bonnet and get a clear view of its workings. Section **??** already dealt with a number of the key concepts in utility theory that the **core model** uses. Where relevant, those will be added to here. The **CES** function and love of variety are the essential components that will be continually returned to.

This section attacks the problem in reverse, starting with the three simultaneous equations toward the endpoint of the analytic argument, which will be called the '**region equations**'. These equations are the 'operational' elements - that is, they can be used to produce simulation code to run the numerical model calculation. In a sense, much of what precedes them is scaffold: once they are reached, a great deal of that scaffold has been analysed away. So, the top-level structure is discussed first, followed by a more detailed look at the scaffold that supports it.

Another reason for taking this approach is that there is no point in merely recreating Brakman *et al*'s thorough step-by-step explanation (which runs for a hundred pages from p.81 to p.182). This section will, however, provide enough information to be able to run a code version of the **core model** and to understand how the components all fit together.

As Neary notes, the **core model** can end up "degenerating at times into a near-impenetrable soup of CES algebra" (Neary 2001 p.537). The goal here is to get to the crux points of the model as clearly as possible first, before dealing with that soup. Though it cannot be completely avoided, much of it can be compartmentalised (as Brakman *et al* do with a series of technical derivations for model subcomponents). It can be particularly confusing trying to locate exactly where the economic mechanisms are, and indeed even where the 'stuff' being consumed is. Often this is because the **GE** approach to model-building is (as with much of neoclassical economics) built on a mix of verbal argument and algebraic rearrangements. Keeping up with these can be like trying to keep one's eye on a ball being moved under several cups.

This will become important when attempting to think through how the model maps to reality and how it compares to an agent approach. It also helps in examining exactly what types of model assumption have been used in the process of putting the **core model** together, and how

these assumptions - both mathematical and verbal - build up the scaffold that supports the final analyses. This knowledge will then be used to compare to a specifically agent-based approach to building a spatial economic model, as well as comparing to Fowler's work on his agent version of the **core model**.

Core model: structure

The **core model** defines **separate regions**, each of which is a dimensionless point, and with transport costs set between them. There are also:

- **Two types of workers**: manufacturing and agricultural. The number of each is set exogenously.
  - **Agricultural workers** are immobile - they cannot move between regions.
  - **Manufacturing workers** *can* move between regions, seeking the best wage.
- **Two types of goods matching the workers**:
  - **Agricultural goods** are a static base for the **core model** where prices cannot change. There are important justifications for this, but these are left to the discussion of market structure (section x) since they can be distracting if trying to get a clear overview.
  - **Manufactured goods**, along with manufacturing workers, are the dynamic part of the model. A single firm makes one variety. The range of varieties produced is the key determinant in workers' utility, through their **love of variety**. The sector has economies of scale (though the somewhat murky nature of these in the **core model** is something that will be discussed in detail). Again, see the discussion on market structure.
- **Manufacturing firms**, who can also move between regions. (As discussed below, however, firms are in reality assumed to be the mirror of workers, and vice versa: they don't make separate decisions or appear explicitly in the model.)

point about firms being explicit: needs tying to Fowler discussion of nature of firms. Point here about how spread out the core model discussion is. Should it all be bought together? Perhaps not, but a guide to where the core model bits are discussed should be given, explaining how the preceding explanations link them.

Much ref to discussion of market structure. Where am I going to put it? It probably should go somewhere closer to the core model outline.

Each region has its own set of the three 'region equations' - these are given in table 1.1. This section deals with the simplest two region version of the model, and so will define the region

region 1	region 2
$Y_1 = \lambda_1 \delta W_1 + (\frac{1}{2})(1-\delta)$	$Y_2 = \lambda_2 \delta W_2 + (\frac{1}{2})(1-\delta)$
$I_1 = \left[\lambda_1 W_1^{1-\varepsilon} + \lambda_2 T^{1-\varepsilon} W_2^{1-\varepsilon}\right]^{1/(1-\varepsilon)}$	$I_2 = \left[\lambda_1 T^{1-\varepsilon} W_1^{1-\varepsilon} + \lambda_2 W_2^{1-\varepsilon}\right]^{1/(1-\varepsilon)}$
$W_1 = \left[Y_1 I_1^{\varepsilon-1} + Y_2 T^{1-\varepsilon} I_2^{\varepsilon-1}\right]^{1/\varepsilon}$	$W_2 = \left[Y_1 T^{1-\varepsilon} I_1^{\varepsilon-1} + Y_2 I_2^{\varepsilon-1}\right]^{1/\varepsilon}$

Table 1.1: 'Regional equations' for two-region core model

equations separately for each of the two regions (as Brakman *et al.* do, p.119; the n-region versions are on p.135).

Before coming on to an explanation of the parameters, what do the 'region equations' achieve? Krugman's question for the **core model** was mentioned in the introduction: "why and when does manufacturing become concentrated in a few regions, leaving others relatively undeveloped?" (Krugman 1991b, p.484). In the model, a **difference in real wages between regions** is the mechanism for answering this: workers are attracted to where wages are higher. There are *two equilibrium stages* to the model. The first ('short-run') finds the **nominal wage in each region**: this is the job of the 'region equations'. This nominal wage can then be used to work out each region's real wage (see below). This sets up a tension to be resolved in the second ('long-run') equilibrium stage: workers move to regions with a higher real wage until regions end up with the same real wage, or all workers move to one region, leaving the other with zero real wage (and thus unable to attract anyone). A possible source of confusion at this point: where have the firms gone? In the **core model**, it is assumed they go where workers go. This will be discussed in detail below.

is this Krugman quote actually mentioned in the introduction?

First mention of confusion about worker/firm duality in core model?

The following explains how all this is defined mathematically. To start with, these are the main variables in the 'region equations' from table 1.1:

- Y: total income, from manufacturing and agricultural workers.
- I: the price index - the 'real' value of manufactured goods, or "what utility can I buy, given the price of manufacturing goods and my **love of variety**?". Its full meaning needs a detailed explanation; this is done below.
- W: The nominal wage rate (not the real wage; see below).

The four parameters used to find those values are as follows. These are all set exogenously, though as will be explained, they can be parameter-swept to understand the model dynamics:

- δ (delta): the share of income spent on manufacturing, as well as the fraction of workers in the manufacturing sector.[1] Note in the equations for total income (Y_1 and Y_2) the term $(\frac{1}{2})(1-\delta)$: this gives the number of agricultural workers in each region. The total number of workers is assumed to be 1 and so the number of agricultural workers is $1-\delta$. They are then split evenly between the two regions.

- λ (lambda): the fraction of manufacturing workers in each region. Note that $\lambda_1 + \lambda_2 = 1$. This means that the number of manufacturing workers in the each region is $\lambda_1\delta$ and $\lambda_2\delta$.

- ε (epsilon): elasticity of substitution; as explained below, this relates to ρ and love of variety, but plays a broader role in the model.

- T (tau): transport costs between regions; the cost of goods is multiplied by this. If a good is being sold where it is produced, transport costs would be 1. This represents there being no internal costs for moving goods. (In the regional equations above, T is thus left out for the home region, since 1 raised to any power is 1). T is defined by Brakman *et al*. as "the number of goods that need to be shipped to ensure that one unit arrives." (p.107) This use of 'iceberg' transports is, again, discussed in more depth below.

Ultimately, the goal of the core model is to find out where mobile manufacturing labour will migrate to. Will it agglomerate in one region - for example, $\lambda_1 = 1$ and $\lambda_2 = 0$? If all workers are in one region, the wage there will be equal for everyone. Or will it spread across the regions, perhaps evenly, where λ_1 and λ_2 both $= 1/2$ and the real wage is equal in both regions? As mentioned, solutions to the regional equations do not supply this answer. Instead, they find the short-run equilibrium: how real wages in each region differ, given exogenous values for λ, δ, ε and T. The regional equations only supply the nominal wage - the real wage is as follows (where r indicates region; the derivation of this is explained below):

$$w_r = \frac{W_r}{I_r^{\delta}} \tag{1.1}$$

As can be seen from table 1.1, the equations are inter-dependent. Intuitively enough, income (Y) is dependent on wages (there are no other sources of income in the **core model** apart from wages). The price index I for manufactured goods is also dependent on wages since - as will be explained below - I roughly translates as "what utility can I buy for my wage?" Finally, the wage itself (W) is dependent on both the price index and income. Ultimately it is the wage that is of interest, as this will determine migration between the regions - but how to solve for it? The answer is: numerically, due to the complexity of the equations[2]. Fujita *et al* set out

[1] It is not obviously the case that these two values should be same; Charles van Marrewijk provides an analysis on the website to accompany the Brakman *et al* book. This can be found at: http://www2.econ.uu.nl/users/marrewijk/newgeo/answers/newgeo%20answer%203.pdf

[2] Extensions to the core model were mentioned above; one of these makes some alterations that allow for a solveable version of the model, but require getting into quite different basic core model structures. See Brakman *et al*. 2009 pp.151

a simple explanation of their numerical method in a footnote[3]; Brakman *et al* expand on this (p.137). It is based on the classic method of guessing a start value, feeding this into the system of equations and iterating until a stable value emerges (for some given definition of 'stable' based on how much the value changes between iterations). Thus, if a guess is taken for W, both Y and I can then be calculated - and these can then be used to re-calculate W until a desirable level of stability is reached[4].

'classic method' of numerical equation system solving - in this case, not Newton's method, I don't think, but need to check.

Once this stable result is found, the real wage can then be worked out. This is the short-run equilibrium, in which real wages between regions can differ. The **core model** can then use those values in its next stage to examine wage migration. The migration stage is actually very straightforward in comparison: the number of manufacturing workers in each region (as determined by the λ fractions) simply flows to where the real wage is higher. (The migration equation used to do this is dealt with further below.)

Where is the migration equation dealt with?

By starting with a top-down view of the structure, it is hoped that the central dynamic of the **core model** is clear, as well as something of the way in which equilibrium solutions are reached. The information supplied above is actually enough to produce a working version of the model; one such version has been coded for the thesis and is described in appendix x, along with a link to an online version with a download link for code included. However, more needs to be said about the scaffold, since important details about the **core model** mechanism relevant to the thesis remain to be explored.

Core model: scaffold

Looking at table 1.1, a number of questions can be asked. Where are the utility functions? Where are the workers? Where are the firms? Where are the economies of scale? How are firms and workers connected - how does the labour market clear? How is firm production consumed?

do I answer this list of questions?

Returning to the absolute basics of the **core model**, utility is defined in a particular way in order to achieve two things. Firstly, utility for the model as a whole derives from a Cobb Douglas function, formulated as described in section **??** and equation (**??**):

$$U = F^{1-\delta} M^{\delta}, (0 < \delta < 1) \tag{1.2}$$

[3] It is very brief: "although it is not necessarily the most efficient procedure, simple iteration generally works: that is, start with guesses at W_r, calculate in sequence the implied Y and I vectors; calculate new values of W; and repeat until convergence" (p.77).

[4] It should be noted, this is skipping over a great many questions about showing that a stable solution is even possible for this system, but this discussion is beyond the scope of the thesis. It will be assumed that known solutions exist and the algorithm can reach them.

1-delta is different. Go back and en-consistemanate.

The **core model** uses a **representative agent**: this one utility function stands in for **all** actors' utility. A vital feature of the Cobb Douglas for the purposes of the **core model** is that demand for F and M are completely separate from each other. As the budgets optimisations of the Cobb Douglas show (again in section **??**), the equations describing the optimal amount of either F or M only contain their own price. This means there is no interdependence between them: changing the price of F can have no effect on demand for M and vice versa. The demand split between F and M is completely controlled by δ, and this shows up in the optimised budget equations. If the price of one of the goods drops, demand rises in exact proportion, keeping the spend on that sector exactly the same. Goods in each of the two sectors are 'unit elastic'. This is a feature of the Cobb Douglas that the **core model** relies on, as it allows M to be optimised separately, knowing that the optimisation is self-contained and cannot effect demand for F. (Section **??** looks at this point in more depth, as it has implications for agent decision-making.)

Need some more detail on elasticity. I think that belongs in Ch.2. Put in about unit elastic; link to why it becomes important.

Whereas F is straightforwardly the quantity of food, M is actually *the utility from manufactured goods*, not the quantity consumed. Intuitively, it is obvious that a single value M could not represent a range of different manufactured goods directly, but this is a potential source of confusion, since - for example in Brakman *et al* - it is sometimes described as straightforward "consumption of manufactured goods" (p.90). Fujita *et al*. (p.47) label M a 'composite index of the consumption of manufactured goods' (ibid p.46) but also note it can "can be thought of as a utility function". To keep things as simple as possible, that is exactly how it will be treated here. The actual mathematical definition of M is a **CES** function containing each variety of consumed manufactured good, of the form described in equation (**??**) and repeated here. N is the total number of varieties consumed; $c(i)$ is the quantity of a single variety:

$$M = \left(\sum_{i=1}^{N} c_i^{\rho} \right)^{\frac{1}{\rho}}, (0 < \rho < 1) \tag{1.3}$$

So, for a given quantity and number of goods, increasing **love of variety** (as $\rho \to 0$) increases M. As will be explained, however, the number of varieties is endogenous in the **core model**.

CES in coremodel: I should explain in ch.2 about how it turns either extra quantity or variety number into extra utility.

The budget constraint for the **representative agent** consuming food and manufactures is as follows, where again $c(i)$ is a single variety and $p(i)$ its cost:

$$F + \sum_{i=1}^{N} p_i c_i = Y \tag{1.4}$$

The separation imposed by the Cobb Douglas applies to any pair of terms in the budget constraint, so as well as knowing that $F = 1 - \delta Y$, it is also the case that $\sum_{i=1}^{N} p_i c_i = \delta Y$. This second term can be used as the budget constraint for M - see below. Note that the price of F cannot change endogenously in the model, and so setting its price to one (leaving just F in the budget constraint) keeps things simple and means F can be used as the numeraire. What else is the agricultural sector achieving in the model? Its job is to provide demand in both regions so that it does not collapse to a point if and when manufacturing workers all move to one region. As mentioned above, in the region equation for Y, each region has $(\frac{1}{2})(1-\delta)$ agricultural income. It is $1-\delta$ across both regions, and this cannot change.

why do I still feel like I'm missing something about the numeraire? Partly: what happens if you remove from model? Need to nail down. And seems like an odd place to discuss the point of agri in the model.

One of the three 'region equations' above is the 'price index' I. This serves a very useful simplifying role in the **core model**. M has already been defined above as 'utility from manufactures' using a **CES** utility function. When M is optimised by its budget constraint ($\sum_{i=1}^{N} p_i c_i = \delta Y$) it becomes possible to rearrange the optimisation[5] to find that $M = \frac{\delta Y}{I}$, where:

$$I = \left(\sum_{i=1}^{N} p_i^{\frac{\rho}{\rho-1}} \right)^{\frac{\rho-1}{\rho}} \tag{1.5}$$

I is a function of the price of each variety, but also contains ρ, the **love of variety** parameter. As Brakman *et al* say, the price index is a "mirror image" of M (p.99) where more varieties - or higher **love of variety** - will *lower* I. Noting again that $M = \frac{\delta Y}{I}$, it can be seen that lowering I increases utility from manufacturing, as would be expected if either variety number or **love of variety** were increased. I thus roughly translates as "what utility can I buy, given the price of manufacturing goods and my **love of variety**?" This definition of I thus simplifies the optimal quantity of M to the budget for manufacturing varieties (δY) and the price index I, and nothing else.

The equation for real wages ((1.1), $w_r = \frac{W_r}{I_r^{\delta}}$) can now be explained in more detail. If W is the nominal wage, the real wage is 'what utility I can buy with W'. (Note, the real wage here is different from the more common understanding of the term, which would be 'what quantity of goods I can buy with my money wage'.) For the **core model**'s whole economy, Fujita *et al* use a 'cost of living index' (p.48; see Brakman *et al* p.99 also) to describe what it costs to buy a unit of utility: $I^{\delta} p_f^{1-\delta}$, where p_f is the cost of food. δ sets the spend on manufactures and so I^{δ} separately describes the cost of a unit of utility from manufacturing. As the numeraire,

[5] Brakman *et al* (p.94-5) and Fujita *et al* (p.47) piece their logic together slightly differently but (apart from the latter using integrable continuous varieties, not discrete) they arrive at the same place.

the price of F is set in the model to one and, since $1^x = 1$, the 'cost of living index' for the whole model reduces to I^δ. Real wages are then simply the nominal wage divided by this 'cost of living index', giving total utility. Differences in utility between regions can then be used to determine where mobile workers will be attracted to.

need to go back and make sure I clearly argue the numeraire point somewhere.

On cost of living index: I'm unsure whether to write up that F being 'stuff' and M being 'utility' is confusing. Mull.

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Todo list

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